

Vector Differentiation: Scalar and vector fields, Gradient, directional derivative, divergence, curl-physical interpretation; solenoidal and irrotational vector fields-illustrative problems.

7+2=9 Hours

1. If $\phi = x^3 + y^3 + z^3 - 3xyz$. Find the $\nabla\phi$ and magnitude of $\nabla\phi$ at $(1, -1, 2)$
2. If $\phi(x, y, z) = 3x^2 + z^2$. Find the $\nabla\phi$ and magnitude of $\nabla\phi$ at $(1, -2, 1)$
3. Find the unit vector normal to the surface $\phi = x^3 + y^3 + z^3 + 3xyz$ at the point $(1, 2, -1)$
4. Find the angle between the surface $9 = x^2 + y^2 + z^2$ and $3 = x^2 + y^2 - z$ at the point $(2, -1, 2)$
5. Find the angle between the surface $x \log z = y^2 - 1$ at $(1, 1, 1)$ & $(-2, -1, 1)$.
6. Find the directional derivative of $\phi = x^2yz + 4xz^2$ at the point $(1, -2, -1)$ in the direction of vector $2i - j - 2k$
7. Find the directional derivative of $\phi = xy + yz + zx$ in the direction of vector $i + 2j + 2k$ at $(1, 2, 0)$
8. Find the directional derivative of $f(x, y, z) = xy^2 + yz^3$ at the point $(2, -1, 1)$ in the direction of vector $i + 2j + 2k$.
9. What is the directional derivative of $\phi = xy^2 + yz^3$ at the point $(2, -1, 1)$ in the direction of normal to the surface $x \log z - y^2 = -4$ at $(-1, 2, 1)$?
10. If $f = \nabla(x^3 + y^3 + z^3 - 3xyz)$ find i) $\nabla \cdot f$ ii) $\nabla \times f$
11. Evaluate i) $\nabla \cdot f$ ii) $\nabla \times f$ at the point $(1, 2, 3)$ where $f = x^2yzi + xy^2zj + xyz^2k$
12. Evaluate i) $\nabla \cdot f$ ii) $\nabla \times f$ at the point $(1, 2, 3)$ where $f = 3x^2i + 5xy^2j + 5xyz^3k$
13. If the vector field $\vec{A} = (ax + 3y + 4z)i + (x - 2y + 3z)j + (3x + 2y - z)k$ is solenoidal find the constant a.
14. Show that the vector field $\vec{F} = (y^2 - z^2 + 3yz - 2x)\hat{i} + (3xz + 2xy)\hat{j} + (3xy - 2xz + 2z)\hat{k}$ is solenoidal.
15. Find the constants a, b, c so that $f = (x + 2y + az)i + (bx - 3y - z)j + (4x + cy + 2z)k$ is irrotational.
16. Find the constant a if the vector $(ax^2y + zy)i + (xy^2 - xz^2)j + (2xyz - 2x^2y^2)k$ has zero divergence.
17. Show that the vector field given by $\vec{A} = (x^2 + xy^2)i + (y^2 + x^2y)j$ is irrotational and hence find the scalar potential.
18. If $\vec{A} = (3xz^2)i - (zy)j + (x + 2z)k$ find $\text{curl}(\text{curl } \vec{A})$.
19. Show that the vector field $\vec{A} = (\sin y + z)i + (x \cos y - z)j + (x - y)k$ is irrotational.
20. Show that the vector field $\vec{A} = (2xyz^3)i + (x^2z^3)j + (3x^2yz^2)k$ is irrotational and find the scalar potential ϕ such that $\vec{A} = \text{grad } \phi$
21. Show that $\text{div}(\text{grad } r^n) = n(n+1)r^{n-2}$, where $r^2 = x^2 + y^2 + z^2$